Randall R. Van Why Jr.

Visiting Assistant Professor, Georgia Institute of Technology

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Education

Northwestern University — Evanston, IL
PhD. Mathematics — *2017 – 2024*

Princeton University — Princeton, NJ
PhD. Fellow — *2021 – 2023*

University of Chicago — Chicago, IL
MSc. in Physical Sciences — *2016 – 2017*

New Mexico Institute of Mining and Technology (NMT) — Socorro, NM
BSc. in Mathematics and Computer Science; Summa cum laude — *2012 – 2016*

Interests: Low Dimensional Geometry and Topology
Algebraic, Complex, Contact, Kähler, Symplectic, Riemannian

Affiliation

Georgia Institute of Technology — Atlanta, GA
Visiting Assistant Professor — *2024 – Present*

Institute for Advanced Study — Princeton, NJ
Visiting Graduate Student — *2019*

Research Articles

Kähler-Ricci Flow is Symplectic Deflation plus an Exact Symplectic Deformation
arXiv.2610.24825 — *Submitted*

Non-Affine Stein Manifolds and Normal Crossing Divisors
arXiv.2507.22290 — *Submitted*

Stein Compactifications and Lefschetz Fibrations
Joint w/ Sierra Knavel-Runner — *In preparation*

Conferences and Seminars

Organization

GT-UGA Joint Geometry Topology Seminar — Atlanta, GA
Georgia Institute of Technology — *2024 – Present*

AMS Eastern Sectional at Clemson University — Clemson, SC
Special Session on Invariants in Geometric Topology: Low Dimensions and Beyond — *March 2025*

Tech Topology Conference — Atlanta, GA
Georgia Institute of Technology — *2024-2026*

Participation

1st Proof Project: Batch 3 Workshop — Pittsburgh, PA
Carnegie Mellon University — *October 2026*

2026 Georgia Topology Conference and Workshop — Athens, GA
University of Georgia — *May 2026*

AMS Southeastern Sectional — Savannah, GA
Georgia Southern University — *April 2026*

2025 Tech Topology Conference — Atlanta, GA
Georgia Institute of Technology — *December 2025*

Midwest Panorama of Geometry and Topology — Iowa City, IA
University of Iowa — *June 2025*

2025 Georgia Topology Conference — Athens, GA
University of Georgia — *May 2025*

Tech Topology Conference — Atlanta, GA
Georgia Institute of Technology — *December 2024*

Workshop in Geometric Topology — Grand Rapids, MI
Calvin University — *June 2024*

Graduate Student Topology/Geometry Conference — East Lansing, MI
Michigan State University — *April 2024*

Tech Topology Conference — Atlanta, GA
Georgia Institute of Technology — *December 2023*

Trisectors Workshop 2023 — Davis, CA
University of California Davis — *June 2023*

2023 Georgia Topology Conference — Athens, GA
University of Georgia — *May 2023*

Midwest Geometry Conference 2023 — Manhattan, KS
Kansas State University — *March 2023*

SYNC Early Career Workshop — Davis, CA
University of California Davis — *August 2022*

Deformations of Geometric Structures in Current Mathematics — New York City, NY
Columbia University — *May 2022*

Graduate Student Topology/Geometry Conference — Online
Indiana University — *April 2021*

Eighth Pacific Rim Conference in Mathematics — Online
University of California Berkeley — *August 2020*

Southern California Geometric Analysis Summer School — Irvine, CA
University of California Irvine — *February 2020*

Summer Northwestern Analysis Program (SNAP) — Evanston, IL
Northwestern University — *August 2019*

Graduate Summer School on the Homological Algebra of Mirror Symmetry — Toronto, Canada
Fields Institute — *July 2019*

Current Trends in Symplectic Topology — Montreal, Canada
Centre de Recherches Mathématiques — *July 2019*

Kylerec 2019 — Truckee, CA
Graduate Workshop — *June 2019*

Tsyganfest — Evanston, IL
Northwestern University — *May 2018*

UChicago/Northwestern Geometric Analysis Day — Evanston, IL
Northwestern University — *April 2019*

Graduate Student Topology and Geometry Conference — Champaign-Urbana, IL
University of Illinois at Urbana-Champaign — *March 2019*

Virginia Topology Conference — Charlottesville, VA
University of Virginia — *December 2018*

MSRI Summer Graduate School: The h-Principle — Tokyo, Japan
University of Tokyo — *June/July 2018*

Graduate Student Topology/Geometry Conference — Chicago, IL
University of Illinois at Chicago — *April 2018*

Geometric Analysis Day 2018 — Chicago, IL
University of Chicago — *March 2018*

Southern California Geometric Analysis Seminar — Irvine, CA
University of California Irvine — *January 2018*

2017 Midwest Geometry Conference — Manhattan, KS
Kansas State University — *November 2017*

Summer Northwestern Analysis Program (SNAP) — Evanston, IL
Northwestern University — *June-July 2017*

Thematic Program on Kähler Geometry — Notre Dame, IN
University of Notre Dame — *June 2017*

UChicago Geometric Analysis Learning Seminar — Chicago, IL
University of Chicago — *October 2016*

University of Chicago Geometry and Topology Summer School — Chicago, IL
University of Chicago — *August 2016*

University of Chicago REU — Chicago, IL
University of Chicago — *June 2015*

AMS Sectional Meeting — Albuquerque, NM
University of New Mexico — *April 2014*

Talks

Conference Talks

AMS Southeastern Sectional at Georgia Southern University — Savannah, GA
Symplectic Divisors and Lefschetz Fibrations in Dimension 4 — *April 2026*

2025 Tech Topology Conference — Atlanta, GA
Divisors and the Geometry of Symplectic 4-Manifolds — *December 2025*

Midwest Panorama of Geometry and Topology — Iowa City, IA
Symplectic geometry, surface configurations, and non-algebraic manifolds — *June 2025*

41st Workshop in Geometric Topology 2024 — Grand Rapids, MI
Symplectic Stories from Algebraic Geometry — *June 2024*

Graduate Student Geometry Topology Conference — East Lansing, MI
Symplectic and Contact Geometries in Low Dimensions — *April 2024*

2023 Tech Topology Conference — Atlanta, GA
Compactifications of Symplectic Manifolds (Cancelled due to illness) — *December 2023*

Trisectors Workshop 2023 — Davis, CA
Divisors and the Trisection of $\mathbb{CP}^2$ — *June 2023*

2023 Georgia Topology Conference — Athens, GA
The Symplectic Geometry of $\mathbb{CP}^2$ — *May 2023*

Kylerec 2019 — Truckee, CA
Liouville and Weinstein Skeleta — *June 2019*

Invited Seminar Talks

Rice University Geometry Seminar — Houston, TX
Non-Affine Stein Manifolds and Normal Crossing Divisors — *October 2025*

The Simons Center for Geometry and Physics Symplectic Seminar — Stony Brook, NY
Non-Affine Stein Manifolds and Normal Crossing Divisors — *October 2025*

University of Alabama Geometry-Topology Seminar — Tuscaloosa, AL
Stein Manifolds and Normal Crossing Compactifications of PALFs — *October 2025*

Columbia University Geometry Seminar — New York City, NY
Non-Affine Stein Manifolds and Normal Crossing Divisors — *September 2025*

University of Georgia Topology Seminar — Athens, GA
Symplectic NC-Divisors, Compactifications, and Non-Affine Symplectic Manifolds — *March 2025*

UMASS Amherst Geometry Topology Seminar — Amherst, MA
Symplectic NC-Divisors, Compactifications, and Non-Affine Symplectic Manifolds — *October 2022*

Seminar Talks

Georgia Tech Analysis Seminar — Atlanta, GA
Kähler-Ricci Flow is Symplectic Deflation Plus an Exact Symplectic Deformation — *Fall 2026 (Planned)*

GT-Geometry Topology Research Seminar — Atlanta, GA
Special Seminar on Spin Geometry — *Summer 2026*

GT-Geometry Topology Research Seminar — Atlanta, GA
Special Seminar on the h-Principle — *Spring 2026*

GT-Geometry Topology Research Seminar — Atlanta, GA
Special Lecture Series on Kähler Geometry (6 lectures) — *Fall 2025*

Special Seminar on The Yang-Mills Equations over Riemann Surfaces — Atlanta, GA
Equivariant Morse Theory — *January 2025*

GT Geometry Literature Seminar — Atlanta, GA
The Structure of Rational and Ruled Symplectic 4-Manifolds by D. McDuff — *November 2024*

GT Geometry Literature Seminar — Atlanta, GA
Morse theory for Lagrangian Intersections by A. Floer — *September 2024*

Georgia Tech Geometry Topology Seminar — Atlanta, GA
Symplectic NC-Divisors, Compactifications, and Non-Affine Symplectic Manifolds — *August 2024*

Northwestern Graduate Student Seminar — Evanston, IL
Toward Compression Progress — *March 2024*

Northwestern Geometry and Physics Seminar — Evanston, IL
Three Lectures on Weinstein/Liouville Geometry — *January 2024*

Northwestern Geometry and Physics Seminar — Evanston, IL
Normal Crossing Divisors and Non-Affine Manifolds — *November 2023*

Northwestern Graduate Student Seminar — Evanston, IL
A tale of Exp and Log — *September 2023*

GT GAPS Seminar — Online
Normal Crossing Divisor Compactifications of Symplectic Manifolds — *April 2023*

Notre Dame Geometry Group Seminar — South Bend, IN
Linear Plumbings of Disk Bundles over Spheres and Symplectic Divisors with Normal Crossings — *July 2021*

Northwestern Geometry Group Seminar — Evanston, IL
Linear Plumbings of Disk Bundles over Spheres and Symplectic Divisors with Normal Crossings — *May 2021*

Northwestern Graduate Student Seminar — Evanston, IL
Milnor's Fibration Theorem — *January 2021*

Notre Dame Graduate Student Seminar — South Bend, IN
Milnor's Fibration Theorem — *December 2020*

Northwestern Geometry Group Seminar — Evanston, IL
Symplectic Divisors and Biran-Giroux Decompositions — *September 2020*

UCLA Special Seminar on Flexibility in Symplectic and Contact Geometry — Los Angeles, CA
Flexibility and Plastikstufe — *July 2020*

Geometry for All Seminar — Evanston, IL
Symplectic Geometry — *March 2020*

Special Symplectic Geometry Seminar — Evanston, IL
Donaldson's Divisor Theorem — *January 2020*

Northwestern Graduate Student Seminar — Evanston, IL
4-Manifolds — *March 2019*

Special Seminar on Gromov's Compactness Theorem for J-Holomorphic Curves — Evanston, IL
Eliashberg's Method of Filling by J-Holomorphic Curves — *March 2019*

Special Seminar on Gromov's Compactness Theorem for J-Holomorphic Curves — Evanston, IL
Gromov's Compactness Theorem — *February 2019*

Special Seminar on Gromov's Compactness Theorem for J-Holomorphic Curves — Evanston, IL
Higher derivative bounds — *January 2019*

New Mexico Tech Philosophy of Mathematics Seminar — Socorro, NM
Various Topics — *2015-2017*

New Mexico Tech ACM — Socorro, NM
Elliptic Curve Cryptography — *February 2016*

New Mexico Tech Topology Seminar — Socorro, NM
The h-Cobordism Theorem — *Fall 2015*

New Mexico Tech Colloquium — Socorro, NM
Configuration Spaces, Braid Groups, and Polynomials — *September 2015*

UChicago REU — Chicago, IL
Configuration Spaces, Braid Groups, and Polynomials — *August 2015*

Teaching

Instruction

Georgia Institute of Technology Math 1554
Linear Algebra — *Instructor*

Georgia Institute of Technology Math 1553
Introduction to Linear Algebra — *Instructor*

Georgia Institute of Technology Math 1551
Single Variable Calculus — *Instructor*

Teaching Assistant/Grading

Northwestern University Math 382
Complex Analysis — *TA/Grader*

Northwestern University Math 344
Abstract Linear Algebra — *TA/Grader*

Northwestern University Math 330
Abstract Algebra — *TA/Grader*

Northwestern University Math 310
Probability Theory — *TA/Grader*

Northwestern University Math 300
Foundations of Higher Mathematics — *TA/Grader*

Northwestern University Math 290
Linear Algebra — *TA/Grader*

Northwestern University Math 285
Accelerated Mathematics for MMSS: Linear Algebra — *TA/Grader*

Northwestern University Math 240
Matrix Algebra — *TA/Grader*

Northwestern University Math 234
Multivariable Integral Calculus — *TA/Grader*

Northwestern University Math 230
Multivariable Differential Calculus — *Exam Grader*

Northwestern University Math 224
Single Variable Integral Calculus — *Exam Grader*

Mentorship

Georgia Institute of Technology DRP — Atlanta, GA
Postdoctoral Mentor — *2025*

Northwestern University DRP — Evanston, IL
Graduate Mentor — *2019,2020*

Northwestern University Mathematics Department — Evanston, IL
Graduate Tutor — *2017-2018*

University of Chicago REU — Chicago, IL
Graduate Mentor — *2016,2017,2018,2019,2020*

Memberships and Sponsors

American Mathematical Society (2014–Present)

Georgia Institute of Technology President's Postdoctoral Fellow (2024–Present)

Princeton University Fellow (2021–2023)

NSF RTG Support DMS-1502632 (2018–2021)

Northwestern University Fellow (2017–2018)